compatibility of the pollen through **recognition-rejection protein reaction** between the pollen and stigma surface. Sexual incompatibility may exist between different species (interspecific) or between members of the same species (intraspecific).The latter is called self-incompatibility. The first visible change in the pollen, soon after it lands on stigma is hydration. The pollen wall proteins are released from the surface. During the germination of pollen its entire content moves into the pollen tube. The growth is restricted to the tip of the tube and all the cytoplasmic contents move to the tip region. The remaining part of the pollen tube is occupied by a vacuole which is cut off from the tip by callose plug. The extreme tip of pollen tube appears hemispherical and transparent when viewed through the microscope. This is called **cap block**. As soon as the cap block disappear the growth of the pollen tube stops.

Pollen tube in the style

After the germination the pollen tube enters into the style from the stigma. The growth of the pollen tube in the style depends on the type of style.

Types of style

There are three types of style a) Hollow or open style b)solid style or closed style c) semi-solid or half closed style.

Hollow style (Open style): It is common among monocots. A hollow canal running from the stigma to the base of the style is present. The canal is lined by a single layer of glandular canal cells (Transmitting tissue).They secrete mucilaginous substances. The pollen tube grows on the surface of the cells lining the stylar canal. The canal is filled with secretions which serve as nutrition for growing pollen tubes and also controlling incompatibility reaction between the style and pollen tube. The secretions contain carbohydrates, lipids and some enzymes like esterases, acid phosphatases as well as compatibility controlling proteins.

Solid style (Closed type): It is common among dicots. It is characterized by the presence of central core of elongated, highly specialised cells called transmitting tissue.This is equivalent to the lining cells of hollow style and does the same function. Its contents are also similar to the content of those cells. The pollen tube grows through the intercellular spaces of the transmitting tissue.

Semi-solid style (half closed type): This is intermediate between solid and open type.

There is a difference of opinion on the nature of transmitting tissue. Some authors consider that it is found only in solid styles while others consider the lining cells of hollow style also has transmitting tissue.

Entry of pollen tube into the ovule: There are three types of pollen tube entry into the ovule(Figure 1.18).

Porogamy: when the pollen tube enters through the micropyle.

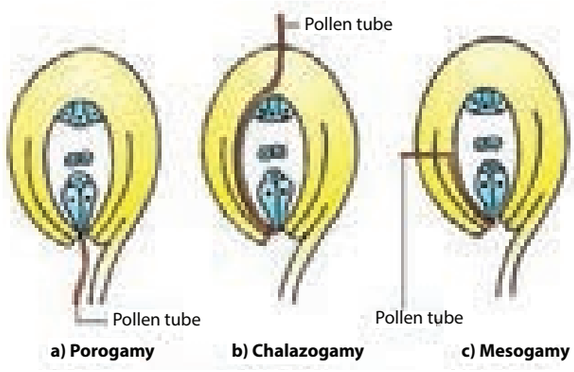


Figure 1.18 Path of pollen tube entry into the ovule

Chalazogamy: when the pollen tube enters through the chalaza.

Mesogamy: when the pollen tube enters through the integument.

Entry of pollen tube into embryo sac: Irrespective of the place of entry of pollen tube into ovule, it enters the embryo sac at the micropylar end. The pollen enters into embryo sac directly into one of the synergids.

The growth of pollen tube towards the ovary, ovule and embryo sac is due to the presence of

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